

Evaluating Domain-Specific Zero-Shot Retrieval without LLMs:

A Comparative Study of CSV-Derived and Benchmark Question Generation

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Abstract

This paper investigates the retrieval performance of embedding-based QA systems in a zero-shot setting without the use of large language models (LLMs). Two distinct evaluation datasets were employed: (1) the *HotpotQA* benchmark (2,000 queries), representing complex multi-hop reasoning; and (2) a *domain-specific university dataset* (10,000 queries), automatically generated from structured CSV metadata of Hanbat National University. Six embedding models—MiniLM, MPNet, MultiMiniLM, and the E5 family (small, base, large)—were evaluated using Recall@5, MRR, and inference latency. The best-performing model, E5-large-v2, achieved a Recall@5 of 0.8750 on HotpotQA but declined to 0.6371 on the university dataset, demonstrating a consistent 20–30% degradation across all models in specialized domains. This finding highlights the limitations of general-purpose embeddings under domain shift and emphasizes the need for domain-adaptive fine-tuning and low-resource alignment methods. **Keywords:** Zero-shot retrieval, sentence embeddings, domain adaptation, HotpotQA, agentic systems.

1 Introduction

Recent advances in Large Language Models (LLMs) have reshaped the landscape of information retrieval and question-answering systems. However, their deployment often entails substantial computational costs, long inference times, and heavy hardware requirements. In contrast, lightweight embedding-based retrieval systems offer efficiency and scalability—critical for agentic systems that must perform autonomous, grounded information retrieval in real time [1].

This work examines the extent to which modern embedding architectures, operating in a zero-shot setting (without task-specific fine-tuning), can generalize to domain-specific retrieval. Specifically, we compare retrieval effectiveness between an established open-domain benchmark (HotpotQA) and a structured domain dataset automatically generated from university information. The study aims to identify key performance gaps and analyze the architectural and dataset-level factors that drive them.

2 Related Work

Dense Retrieval Models. Early dense retrievers such as DPR [2] and Sentence-BERT [3] demonstrated the utility of bi-encoder representations for semantic search. Subsequent work, including MPNet and MiniLM, optimized pre-training objectives for compact and high-performance embeddings.

Contrastive Fine-Tuning Approaches. The E5 series [4] introduced supervised contrastive objectives with instruction-style data, achieving strong zero-shot performance across benchmarks. However, their robustness in highly specialized domains remains underexplored.

Domain Adaptation in Retrieval. Prior studies [5] suggest that embeddings trained on generic corpora exhibit reduced transferability to domain-specific contexts due to vocabulary and entity mismatches. Our work quantitatively examines this degradation using structured, automatically generated university datasets.

3 Methodology

3.1 Dataset Construction

Two datasets were employed:

- **HotpotQA Benchmark:** Adapted from the validation split of the distractor setting ($\approx 2,000$ queries). It involves multi-hop reasoning over open-domain text.
- **University Domain Dataset:** Generated from structured CSV files including course descriptions and professor metadata (total $\approx 10,000$ QA pairs). Each record was converted into factual single-hop question-answer pairs, e.g., “What is the research area of Professor X?” or “What is the course code of Y?”. No manual correction was applied, reflecting realistic data generation noise.

3.2 Evaluation Models and Metrics

We evaluated six models: MiniLM, MPNet, MultiMiniLM, and E5-small-v2, E5-base-v2, E5-large-v2. Metrics used:

$$\text{Recall@5} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\text{relevant} \in \text{Top-5}_i), \quad \text{MRR} = \frac{1}{N} \sum_{i=1}^N \frac{1}{r_i}$$

where r_i denotes the rank of the first relevant document. We also report average inference time per query (ms) to assess latency trade-offs.

4 Results

4.1 HotpotQA Benchmark

Table 1: Retrieval Performance on HotpotQA (2,000 Queries)

Model	Size (M)	Recall@5	MRR	Time (ms/q)
MiniLM	22	0.7300	0.6308	2.95
MPNet	110	0.7665	0.6621	24.90
MultiMiniLM	38	0.4690	0.3663	3.32
E5-small-v2	50	0.8620	0.7960	10.38
E5-base-v2	110	0.8645	0.8038	19.98
E5-large-v2	330	0.8750	0.8200	35.00

4.2 University Domain Dataset

Table 2: Retrieval Performance on Domain-Specific Dataset (10,000 Queries)

Model	Size (M)	Recall@5	MRR	Time (ms/q)
MiniLM	22	0.6441	0.5019	0.52
MPNet	110	0.6116	0.4750	2.03
MultiMiniLM	38	0.5211	0.3719	0.69
E5-small-v2	50	0.6712	0.5335	0.70
E5-base-v2	110	0.6468	0.4996	2.05
E5-large-v2	330	0.6371	0.4931	7.83

5 Discussion

The results demonstrate a consistent decline in retrieval accuracy across all models when applied to domain-specific data. This degradation cannot be fully explained by model scale, as E5-large-v2—with six times more parameters than MiniLM—still experienced a 27% reduction in Recall@5. The likely contributing factors include:

- **Lexical mismatch:** domain-specific entities and terminology absent in training corpora.
- **Sparse contextual cues:** CSV-derived text lacks narrative structure, reducing semantic richness.
- **Noise from automatic generation:** unverified QA pairs introduce retrieval ambiguity.

Thus, purely general-purpose embeddings show limited robustness in structured, non-naturalistic text retrieval.

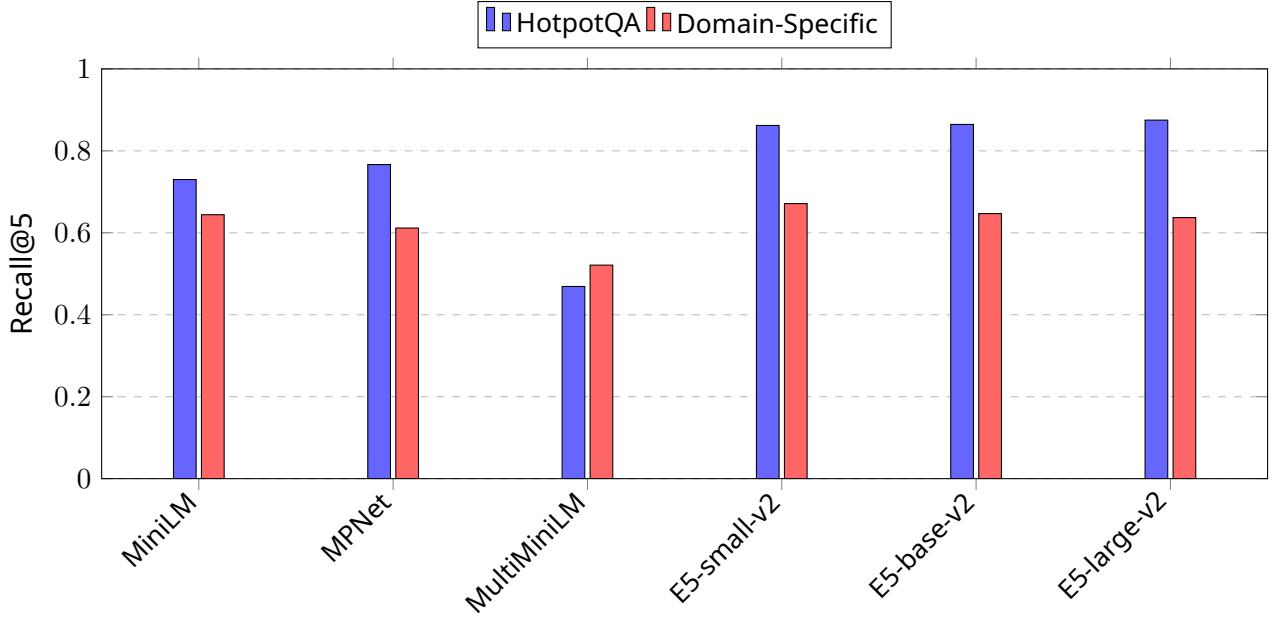


Figure 1: Recall@5 comparison between HotpotQA and Domain-Specific datasets. All models exhibit a 20–30% performance drop under domain shift.

6 Conclusion

This study reveals a persistent performance gap between general and domain-specific retrieval in zero-shot embedding models. Even large-scale architectures like E5-large-v2 fail to maintain benchmark-level performance when transferred to structured data contexts. Future work should focus on domain-adaptive fine-tuning, contrastive representation learning with authentic in-domain corpora, and adaptive negative sampling strategies. Achieving robust domain adaptation in embedding models is essential for enabling efficient, grounded agentic systems in enterprise and academic settings.

References

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